

Day 2

Generative AI & Retrieval-Augmented Generation

Ground every answer in trusted A-001 knowledge - and prove it with sources.

- Practical LLM behaviour: prompts, tokens, context, temperature
- Hallucination risk and the enterprise evidence rule
- Chunking, embeddings and semantic retrieval
- A governed A-001 RAG pipeline with cited answers

Ask a General LLM About A-001

Try it live before you explain anything.

ASK A GENERAL LLM

"Which inspection procedure is currently approved for A-001, and what does it require?"

1 · IT MAY NOT KNOW

Private procedures, current approvals and internal reports are not inside the model.

2 · KNOWLEDGE CHANGES

Procedures are revised. Training data has a cutoff; approvals change after it.

3 · IT WILL STILL ANSWER

A fluent, confident, invented procedure is the most dangerous output of all.

A confident answer is not the same as a correct or authorised answer.

The fix is not a bigger model. The fix is evidence - retrieval plus generation, under authority rules.

Classical ML vs Generative AI

Both learn from data - they solve different kinds of problems.

CLASSICAL ML - YESTERDAY

Input: structured features (vibration, temperature)

Output: a prediction - a number or a class

Strength: measurable, specific, scored honestly

Limit: cannot read procedures or write a briefing

GENERATIVE AI - TODAY

Input: language - instructions and context

Output: new language - answers, summaries, drafts

Strength: reads, reasons and writes across documents

Limit: fluent does not mean grounded or approved

Use ML for measurable prediction. Use GenAI for language work - with evidence attached.

LLM Foundations in Practical Terms

The working vocabulary for everything we build today and tomorrow.

Concept	Practical meaning
Prompt	The instruction and context sent to the model
Token	A unit of text; limits and cost are measured in tokens
Context window	How much the model can see in one interaction
Temperature	Output variability - lower for bounded enterprise tasks
Structured output	A predictable schema (e.g. JSON) for downstream systems
Hallucination	A fluent answer that is unsupported or invented
Foundation model	General-purpose; needs instructions, evidence and controls

An LLM is a general engine. Enterprise value comes from the evidence and controls around it.

Watch: LLM Basics - Generative vs Reasoning

Narrated explainer video - press play

PREDICT THE NEXT TOKEN

LLM basics: one trick, two families, one dial, one hard limit.

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NEXT TOKEN / 02

01 THE ONE TRICK

"The A-001 vibration is █"

rising		0.42
high		0.31
normal		0.18
falling		0.09

Trained on enormous amounts of text. Everything else builds on this one trick.

02 GENERATIVE MODELS

Built to produce fluent text

drafts · summaries · explanations · rewording

fast · cheaper

Your everyday workhorse: given instructions and context, it writes. Most enterprise drafting starts here.

03 REASONING-OPTIMISED

Works step by step first

plans · checks · multi-step logic before answering

slower · more expensive

Same engine, trained further to think before it speaks. Reach for it when the problem has steps, not when it has wording.

04 ONE DIAL - TEMPERATURE

LOW - most likely continuation, steady and predictable

Enterprise answers grounded in evidence: keep temperature LOW.

HIGH - samples widely, more varied, less predictable

05 THE HARD LIMIT

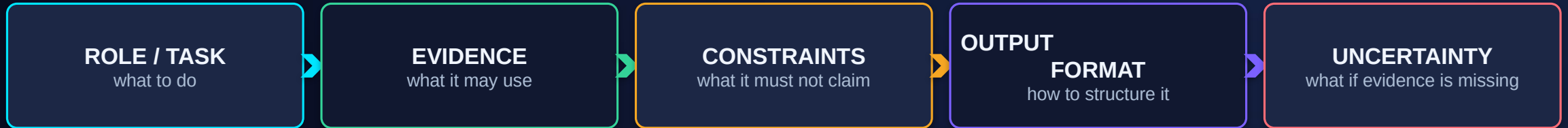
The model knows language, not your documents. That gap is exactly what retrieval fills next.

ILLUSTRATIVE DATA · Synthetic A-001 scenario

Fluent text is not verified fact

The Five-Part Prompting Pattern

Every enterprise prompt in this programme follows the same shape.



A-001 EXAMPLE

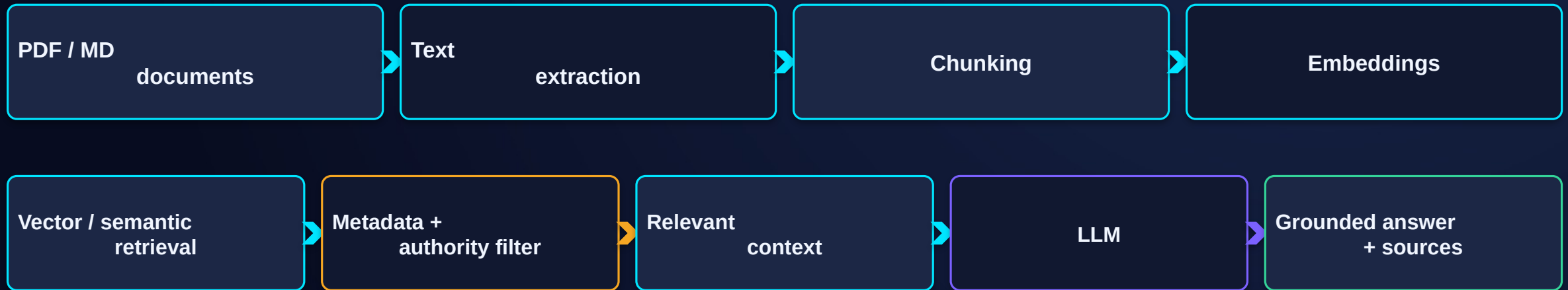
“You are a reliability assistant. Answer **only** from the supplied approved documents. Cite the document ID for every claim. If the approved source is missing, say so - do not guess. Answer in three sentences.”

The uncertainty clause is not optional - it is where hallucination goes to be caught.

Try it: add one part at a time to the opening question and watch the answer change character.

How RAG Works

Retrieve trusted context first - then let the model write from it.



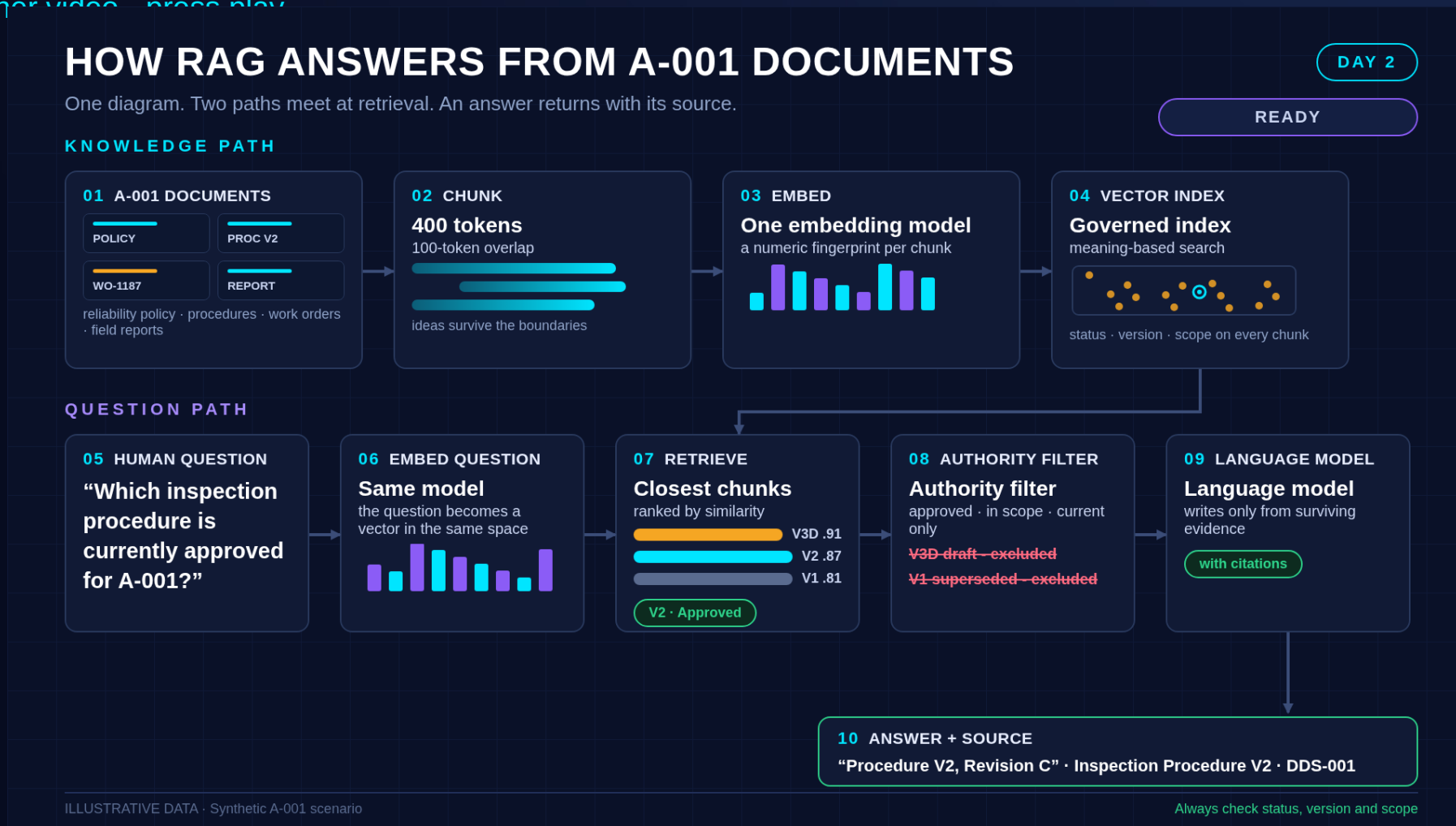
WHY THIS ORDER MATTERS

Retrieval decides what the model is allowed to see. Generation only explains what retrieval found. If the wrong documents reach the model, no prompt can fully protect the answer - which is why the authority filter sits before generation, not after.

RAG = retrieve evidence, apply authority rules, then generate - with sources attached.

Watch: How RAG Works

Narrated explainer video - press play



Watch: Chunking a Real Procedure

Narrated explainer video - press play

CHUNKING A REAL PROCEDURE

Deep dive: A-001 Inspection Procedure V2 becomes retrievable pieces.

01 THE DOCUMENT

A-001 Inspection Procedure

VERSION V2 · APPROVED · 41 PAGES

Retrieval never reads documents - it reads chunks.

02 THE CUTS - 400 TOKENS, 100 OVERLAP

CHUNK 1
scope and definitions

CHUNK 2
preparation - steps 1-4

CHUNK 3
steps 5-7 · acceptance criteria

03 EVERY CHUNK CARRIES METADATA

document	A-001 Inspection Procedure
version	V2
status	APPROVED
scope	A-001

That metadata is what makes governance possible later.

04 THREE THINGS TO REMEMBER

chunks are the unit of retrieval

overlap protects meaning at the boundaries

metadata travels with every chunk

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CHUNK THREE / 05

ILLUSTRATIVE DATA · Synthetic A-001 scenario

Retrieval reads chunks, not documents

Watch: How the Retriever Finds Chunks

Narrated explainer video - press play

HOW THE RETRIEVER FINDS CHUNKS

Deep dive: similarity ranks, authority filters - two different mechanisms.

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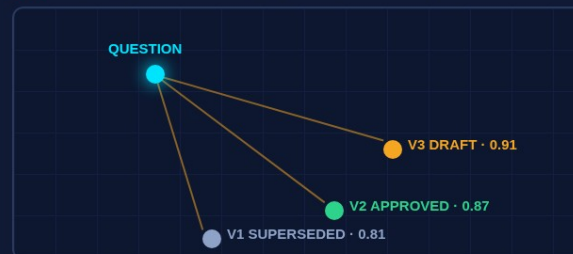
THE PRINCIPLE / 07

01 THE QUESTION

"Which inspection procedure is currently approved for A-001?"

Embedded by the **same model** as the chunks - so the question becomes a **point in the same vector space**.

02 THE VECTOR SPACE - COSINE RANKING



#1 · V3 DRAFT chunk	0.91
#2 · V2 APPROVED chunk	0.87
#3 · V1 SUPERSEDED chunk	0.81

03 THE AUTHORITY FILTER - METADATA

V3 chunk

status: draft

EXCLUDED

V1 chunk

status: superseded

EXCLUDED

V2 chunk

status: approved

SURVIVES

Only approved V2 chunks go to the model ...

04 TWO DIFFERENT QUESTIONS, TWO DIFFERENT MECHANISMS

similarity is measured in the vector space

authority is checked in the metadata

confusing them is how assistants cite drafts

ILLUSTRATIVE DATA · Synthetic A-001 scenario

Relevant is not authoritative

Governed RAG - Relevance is Not Authority

Nine A-001 documents, one approved answer

Synthetic training simulation
Illustrative data - not plant output

1 · THE QUESTION

"Which inspection procedure is currently approved for A-001, and what does it require?"

2 · CHUNK + EMBED

9 A-001 documents → passages
chunk size **400** · overlap **100**
each chunk → embedding vector
question → same embedding model

3 · THE GOVERNANCE RULE

Retrieve by meaning, then filter by **approval status + version + entity scope** before anything reaches the model.

4 · SEMANTIC RANKING - MEANING ONLY

A001-PROC-INS-001-V3D
draft procedure - newest, closest wording

similarity 0.91

DRAFT

A001-PROC-INS-001-V2
approved inspection & PM procedure

similarity 0.89

APPROVED

A001-PROC-INS-001-V1
original procedure - replaced

similarity 0.70

SUPERSEDED

A001-TSG-001
troubleshooting guide

similarity 0.66

APPROVED

↑ ranked by meaning alone, the DRAFT wins. That is the trap.

5 · AUTHORITY FILTER

V3D → **excluded - DRAFT**

V1 → **excluded - SUPERSEDED**

V2 → **eligible - APPROVED, current**

TSG → eligible - approved context

6 · GROUNDED ANSWER

"The currently approved procedure is **A001-PROC-INS-001-V2, Inspection & PM Procedure**. It requires vibration and temperature checks against the V2 limits and recording results in the inspection log. Draft V3D exists but is **not yet approved** and was not used."

sources: A001-PROC-INS-001-V2 · status: APPROVED

WHY IT MATTERS

Newest ≠ authoritative. The model answers only from evidence that survives the filter - and shows its sources.

Relevant is not the same as authoritative. Retrieve evidence, then apply authority rules - then generate.

Chunking and Embeddings, Minus the Maths

Small searchable passages + numeric meaning.

BEGINNER RETRIEVAL SETTINGS

Chunk size 400 - coherent context, still findable

Overlap 100 - ideas survive chunk boundaries

Top K 3 - a small, inspectable evidence packet

EMBEDDINGS - THE MENTAL MODEL

An embedding turns text into numbers so semantically related text can be compared. Similar meaning, similar numbers - even when the exact words differ. You never need to visualise all 1,536 dimensions; you need the comparison.

WHY NOT ONE GIANT DOCUMENT?

Whole documents dilute the match and overflow the context window. Chunks let retrieval return exactly the relevant passage - and let us carry document ID, version and approval status alongside every passage for filtering and citation.

Chunking decides what can be found. Metadata decides what may be used.

Keyword Search vs Semantic Retrieval

Use both deliberately - hybrid retrieval beats either alone.

KEYWORD / LEXICAL

- Best when exact terms matter: A001-PROC-INS-001-V2, part numbers, limits
- Easy to explain and debug
- Misses synonyms and paraphrases

SEMANTIC / VECTOR

- Best when meaning matters: "pump running rough" vs "abnormal vibration"
- Handles natural-language questions
- Can retrieve similar-but-not-authoritative content - the draft trap

Exact IDs favour keyword; natural questions favour semantic. Governed retrieval uses both, then filters by authority.

In the notebooks: a hybrid ranker scores meaning first, then applies approval status, version and entity scope before anything reaches the model.

Nine Documents and the Procedure Trap

The corpus is deliberately messy - like every real enterprise.

Version	Status	Treat as current authority?
Inspection Procedure V1	SUPERSEDED	No - historical context only
Inspection & PM Procedure V2	APPROVED / current	Yes - current procedural authority
Inspection Procedure V3D	DRAFT	No - newer does not mean approved

THE GOVERNANCE TRAP

The newest or most semantically similar document is not automatically authoritative. V3D is newer but DRAFT; V2 remains the current APPROVED procedure. Enterprise retrieval must consider relevance AND authority.

Relevant does not mean trusted. Retrieve evidence, apply authority rules, then generate.

Watch: Relevant Is Not Authoritative

Narrated explainer video - press play

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THE TRAP / 04

RELEVANT IS NOT AUTHORITATIVE

The governance trap, live on one A-001 question.

01 THE QUESTION

“Which inspection procedure is currently approved for A-001?”

asked by an engineer, answered from the controlled corpus

02 RANKED BY MEANING ALONE

PROCEDURE V3D newest document · closest wording DRAFT	0.91
PROCEDURE V2 the approved procedure APPROVED	0.87
PROCEDURE V1 older wording SUPERSEDED	0.81

03 AUTHORITY FILTER · RUNS BEFORE GENERATION

04 THE ANSWER

“Procedure V2 is the currently approved inspection procedure for A-001.”

cites: A-001 Inspection Procedure V2 · refused: V3D (draft)

ILLUSTRATIVE DATA · Synthetic A-001 scenario

Always check status, version and scope

Participant Exercise: Catch the Draft

Six steps from question to cited, governed answer.

01 Ask about the current inspection procedure

One clear A-001 question

02 Inspect the top retrieved chunks

Read the evidence packet + metadata

03 Flag DRAFT or SUPERSEDED material

Find the trap with your own eyes

04 Apply the approval / version rule

Filter before generation

05 Generate with citations

Answer cites the approved source

06 State what stays uncertain

Name gaps needing human review

HANDS-ON Notebooks: [A001_Simple_RAG_LlamaIndex_Databricks.ipynb](#) + [01_A001_Simple_SQL_RAG.ipynb](#)

Evaluate the Answer, Not the Vibe

A grounded answer survives six checks - and fails loudly.

SIX EVALUATION CHECKS

- Did retrieval return a relevant source?
- Did it respect approval status and version?
- Does the answer separate evidence from model wording?
- Are citations and provenance visible?
- Does it avoid inventing facts to fill gaps?
- Does it preserve the human decision boundary?

COMMON FAILURE MODES

No documents found → path / upload / extensions

Wrong procedure returned → metadata + approval filter

Good source, poor answer → prompt structure

No citations → source fields not carried through

Overstated certainty → system prompt uncertainty clause

A RAG system you cannot evaluate is a demo, not a capability.

A Question Documents Alone Cannot Answer

Tomorrow the assistant gains tools - and limits on using them.

THE QUESTION THAT BREAKS RAG

"What was happening in A-001's data, what does the current approved procedure say, and what should be prepared for human reliability review?"

NEEDS STRUCTURED FACTS

Health score, open work orders, sensor trend - live from tables, not documents.

NEEDS APPROVED GUIDANCE

The current procedure and what it requires - from governed retrieval, with citations.

DAY 3 Day 3: one governed agent that chooses the right tool - and stops where its authority ends.